

Logs maintained by the users using dial up connections to connect to ISP indicate the time and duration on Internet usage / connectivity.

It will corroborate other types of evidence that has already been gathered during investigation. Admissibility of such an evidence can not be questioned as it clearly highlights not only the time and duration of computer's Internet usage / connectivity, but also the fact that during a specific time period, the user's computer had been working as a sort of 'computer network'.

Problems with Preserving Computer Evidence

- *Some of the facilities within the browsers to save WWW pages to the disk are imperfect- as it may save the text but not the associated images.*
- (b) In case of some very complex pages, involving "frames" and "templates", there could be perceptible difference in what is seen on screen and what is saved on the disk.*
- (c) It is difficult to tell when a specific page was last acquired, thanks to browser cache facility.*

Thus if one examines a whole series of cached pages, it is not easy to pinpoint which page came first and which later.

Problems with Preserving Computer Evidence

(d) It should not be forgotten that many ISPs use proxy servers to speed up the delivery of popular pages. Thus a user of such an ISP may not be sure that what he has received on his computer is the latest version from the source computer (website) as opposed to an earlier cached version held by his ISP.

Common Computer Forensic Techniques

An understanding of common computer forensic techniques will go a long way to obtain and preserve - authentic, accurate and complete evidence:

- *Carrying out a pre-raid intelligence review to assess what types of hardware, software and back-ups might be expected.*
- (ii) Photographing (video) the computer(s) in situ, particularly any cabling of peripherals and ancillaries,*

Common Computer Forensic Techniques cont..

- (iii) Every computer and its input and output devices carry a product number, it is thus advisable that these numbers be duly recorded*
- (iv) Careful identification and labeling of all items, including cables, peripherals, external data storage medium, like CDs, disks and tapes*
- (v) Precautions to prevent the data being destroyed by hostile individuals immediately prior to the raid,*

Common Computer Forensic Techniques cont..

- (vi) Proper handling of computers that are running, at the time of the raid,*
- (vii) Following proper procedures for safe shutting down,*
- (ix) Noting of the time on the computer's internal clock which provides date and time stamps on computer files*
- (x) Making of an exact sector-by-sector copy* of every hard disk*

Common Computer Forensic Techniques cont..

The method consists of starting (or "booting") the computer not from the hard disk but through a floppy (Drive: A).

The operating system will make an "image" of the hard disk on the external device. The image is an exact copy, also referred as 'bit copy' or 'sector-by-sector copy' of the original.

It includes not only the visible files on the original hard disk but other which would not be seen, the parts of the disk that contain the information from which the directory details are obtained (file names, sizes, date and time stamp) and also certain other forensic segments from previously deleted files can sometimes be recovered.

Credit card fraud

(From Wikipedia, the free encyclopedia)

Credit card fraud is a kind of fraud where a merchant (business, service provider, seller, etc.) is "tricked" into releasing merchandise or rendering services, believing that a credit card account will provide payment for goods/services. The merchant later learns that they will not be paid, or the payment they received will be reclaimed by the card's issuing bank.

Typically, the fraudster causes a credit card of another person to be charged for a purchase. Today, half of all credit card fraud is conducted online, meaning that the fraudsters make online purchases with the credit card details of other people.

Types of Fraud

Stolen Card Fraud

When a card holder loses or has their credit card stolen, it is possible for the thief to make unauthorized purchases on that card up until the card is cancelled. Businesses that accept credit cards are not permitted to request supplemental ID from the cardholder, unless the credit card is not signed. A thief can potentially purchase thousands of dollars in merchandise or services before the card holder or the bank realize that the card is in the wrong hands. Self-serve payment systems such as gas stations are also highly prone to accepting a stolen credit card, as there is no verification of the card holder's identity, however many stations are trying to prevent this by adding a check requiring the user to key in a zip code. The zip code must match the code registered to the credit card or the transfer will fail.

Account Takeover Fraud

Fraud perpetrators call in and impersonate actual cardholders using stolen personal information. They have the address and other information of the cardholder changed to an address they control. Additional cards and possibly PIN mailers are requested and issued to the new address and used by the fraudsters to make purchases and/or obtain cash advances.

Sometimes the fraudster will attempt to add themselves or an alias that they control as an authorized user to the account in order to make it easier to commit the fraud.

Credit Card Mail Order Fraud

Using a stolen credit card number, or computer generated card number, a thief will order stolen goods.

Skimming

Skimming is the theft of credit card information by a dishonest employee of a legitimate merchant, manually copying down numbers, or using a magnetic stripe reader on a pocket-sized electronic device. Common scenarios for skimming are restaurants or bars where the skimmer has possession of the victim's credit card out of their immediate view. The skimmer will typically use a small keypad to unobtrusively transcribe the 3 or 4 digit Card Security Code which is not present on the magnetic strip.

Many instances of skimming have been reported where the perpetrator has put a device over the card slot of a public cash machine (Automated teller machine), which reads the magnetic strip as the user unknowingly passes their card through it. These devices are often used in conjunction with a pin-hole camera to read the user's pin number at the same time.

Cards in countries such as the UK are issued featuring a smart chip with public key encryption. The chip cannot be copied, but the card number, expiry date and security code can be, and this set of data is often sufficient to use the victim's credit card account for fraudulent purposes with so-called "card not present" transactions, e.g., manual input, over the telephone or internet.

Carding

Carding is a term used by fraudsters for a process they use to verify that sets of stolen credit card data are still valid. The fraudster will present each set of credit card details in turn on a website that has real-time transaction processing, making a purchase for a very small monetary amount so as not to use up the card's credit limit, and so as not to attract the attention of a human reviewer to the transaction.

Often, an online donation site for a charity is used instead of an eCommerce merchant, since there is no need to find an item of a suitable price to put in the virtual shopping cart, nor to supply shipping details. The carder may do this manually with a web browser, or may write automated software to interface to the website's checkout or billing forms.

In the past, carders used to use computer programs called "generators" to produce a sequence of credit card numbers, and then test them to see which were valid accounts. However, this process is no longer viable due to widespread requirement by internet credit card processing systems for additional data such as the billing address, the 3 to 4 digit Card Security Code and/or the card's expiry date. Nowadays, carding is more typically used to verify credit card data obtained directly from the victims by Skimming or Phishing.

A set of credit card details that has been verified in this way is known in fraud circles as a phish (see Phishing). A carder will typically sell data files of phish to other individuals who will carry out the actual fraud. Market price for a phish ranges from US\$1.00 to US\$50.00 depending on the type of card, freshness of the data and credit status of the victim.

Credit Card Crime Profits, Losses & Punishment

Losses

U.S. Federal Law can hold the cardholder victim responsible for up to \$50. Merchants in high-risk industries, like unattended automated fuel pumps or Internet sales, anticipate a certain amount of credit card fraud, and set prices accordingly. These higher costs are then passed onto the customer.

Credit Card Companies

In 2003 the Wall Street Journal estimated that the credit card industry generated \$500,000,000 in annual revenue in research and investigation fees paid by consumers and businesses. This additional revenue offsets some of the costs incurred by credit card issuing and processing companies' when investigating chargeback claims. Some merchants believe the high revenue generation by the banks from the crime victims, reduce the incentive for the credit card banks to implement procedures to reduce credit card crime. However, the companies which collect these fees are not capable of dictating fraud prevention policies to the rest of the world. Payment transfer associations, like Visa and Mastercard, receive profit from transaction fees calculated as a percentage of the amount of money they transfer. These associations are motivated to enact policies which increase the amount of money transferred by their systems. Credit card fraud has a chilling effect on merchant acceptance of credit cards, motivating merchants to not accept credit card payments to mitigate their risk of loss. These payment transfer associations are therefore motivated to enact policies and enforce regulations which reduce credit card fraud.

Merchants have begun to request changes in State and Federal Laws to protect consumers and merchants from fraud, but the credit card industry has opposed many of the requested laws.

Because all card-accepting merchants and card-carrying customers are bound by contract law, according to the agreements they sign with their processing / issuing banks, respectively, State and Federal law has a smaller role in preventing merchants from being tricked. Payment transfer associations enact regulatory changes, and issuing / acquiring banks, merchants, and cardholders are contractually bound to these new regulations.

The Criminals

In the US, persons that commit credit card crime largely go unpunished and repeatedly victimize consumers and businesses. The Secret Service handles crimes involving the US money supply, they have a limit of \$2,000 before investigating each crime. Most credit card criminals know this and keep purchases from any one business below \$2,000. With credit card crime occurring across state lines, criminals often are never prosecuted because the

dollar amounts are too low for local law enforcement to pay for extradition.

Reporting Credit Card Fraud

If you lose or have had your credit card stolen, you should immediately report it to your card issuer. Once you report the incident, you are no longer responsible for unauthorized charges made on your card.

In the US, credit card fraud can be reported to the Federal Trade Commission (FTC) and to local and regional authorities. It is the standing policy of the FTC not to investigate reports where the value of fraud does not exceed \$2000. Local law enforcement may or may not further investigate a credit card fraud, depending on the amount, type of fraud, and where the fraud originated from.

Tips to help you avoid credit card fraud

- Don't carry credit cards you don't use and never leave them unattended in a purse, briefcase or wallet.
- Always make sure you get your credit card receipt because it just may include your credit card number. Never toss it in a public trash bin. You'll need that receipt later anyway to tally things up when your statement arrives.
- Shred all documents that might include your credit card number before disposing of them - old slips, credit card statements, bills, anything.
- Never give your number over the phone to someone that you do not know. It's OK if you initiate the call but if you get a call at home from anyone that you do not know by name do not give them your credit card number.
- Never respond to an e-mail asking for your number, no matter how official or legitimate it looks. These bogus e-mails are the #1 fraud right now on the Internet.
- Review your monthly statement as soon as it comes and report any problems right away. To insure your rights, follow-up by filing a written complaint form.
- If using a bank card where you have the option to use credit or debit, ALWAYS USE THE CREDIT OPTION as this may prevent the skimmer device from accessing your four digit pin number.
